through the population. As a consequence of this some benefit would start to accrue to individuals who did not give up immediately, but waited for a few seconds before giving up. This strategy would pay when played against the immediate retreaters who now predominate in the population. Selection would then favour a progressive extension of the giving-up time until it once more approached the maximum allowed by the true economic worth of the resource under dispute.

Once again, by using words, we have talked ourselves into picturing an oscillation in a population. Once again, mathematical analysis shows that this is not correct. There is an evolutionarily stable strategy, which can be expressed as a mathematical formula, but in words what it amounts to is this. Each individual goes on for an *unpredictable* time. Unpredictable on any particular occasion, that is, but averaging the true value of the resource. For example, suppose the resource is really worth five minutes of display. At the ESS, any particular individual may go on for more than five minutes or he may go on for less than five minutes, or he may even go on for exactly five minutes. The important thing is that his opponent has no way of knowing how long he is prepared to persist on this particular occasion.

Obviously, it is vitally important in the war of attrition that individuals should give no inkling of when they are going to give up. Anybody who betrayed, by the merest flicker of a whisker, that he was beginning to think of throwing in the sponge, would be at an instant disadvantage. If, say, whisker-flickering happened to be a reliable sign that retreat would follow within one minute, there would be a very simple winning strategy: 'If your opponent's whiskers flicker, wait one more minute, regardless of what your own previous plans for giving up might have been. If your opponent's whiskers have not yet flickered, and you are within one minute of the time when you intend to give up anyway, give up immediately and don't waste any more time. Never flicker your own whiskers.' So natural selection would quickly penalize whisker-flickering and any analogous betrayals of future behaviour. The poker face would evolve.

Why the poker face rather than out-and-out lies? Once again, because lying is not stable. Suppose it happened to be the case that the majority of individuals raised their hackles only when they were truly intending to go on for a very long time in the war of attrition. The obvious counterploy would evolve: individuals would give up immediately when